

Reflection and Traceability Report on Physiotherapy Companion

Team 11, technically functional

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[Reflection is an important component of getting the full benefits from a learning experience. Besides the intrinsic benefits of reflection, this document will be used to help the TAs grade how well your team responded to feedback. Therefore, traceability between Revision 0 and Revision 1 is an important part of the reflection exercise. In addition, several CEAB (Canadian Engineering Accreditation Board) Learning Outcomes (LOs) will be assessed based on your reflections. —TPLT]

1 Changes in Response to Feedback

1.1 SRS and Hazard Analysis

The hazard analysis document was improved through the peer review along with TA feedback in several areas for enhancement, including clarifying the roadmap with implementation timelines and rationales for safety requirements (#144), adding justification for future work prioritization (#150), and removing problematic assumptions about user familiarity with smart devices (#136). Inconsistencies between the SRS and Hazard Analysis were resolved (#129), while the system boundaries and components section was expanded with lower-level explanations to remove ambiguity (#135). The FMEA table was further decomposed to represent detailed failure modes rather than general effects (#142), and hazard categorization was clarified for technical hazards that could impact safety (#153). Additional improvements included adding rationales for each critical assumption (#136), addressing inconsistencies between assumptions and failure modes (#143), and noting where mitigation strategies would be expanded (#152). TA feedback further reinforced the need for roadmap rationales, which was addressed alongside similar peer feedback. One comment regarding bibliography completeness (#151) was determined to be invalid as the citation and reference were already properly included.

Source	Issue #	Issue Description	Changes Made	Corresponding Commit
Peer Review	#151	Citation "IppolitoWallace1995" appears but bibliography incomplete	No change - comment invalid as bibliography and in-text citation were complete	None
Peer Review (Group 14)	#144	Unclear description of how/when safety requirements and hazard testing will be implemented	Added section in roadmap outlining rationale for accepting requirements within capstone timeline and rationales for excluded safety requirements	939b502

Source	Issue #	Issue Description	Changes Made	Corresponding Commit
Peer Review (Group 14)	#150	Safety requirements prioritized as "first" vs "in the future" without clear justification or timeline	Added clear rationale for future work prioritization	470F490
Peer Review (Group 14)	#136	Missing rationale for each assumption; assumption about user familiarity with smart device should be avoided	Removed assumption about user familiarity; added rationales for each assumption	a103812
Peer Review (Group 14)	#143	Inconsistency between assumption about user familiarity and "confusing prompts" failure mode	Removed the initial assumption; will update language of "confusing prompts"	None (different commit)
Peer Review (Group 14)	#129	Inconsistencies between SRS and Hazard Analysis; hazards need prioritization by severity	Updated consistency between SRS and HA	Reference #410 PR
Peer Review (Group 14)	#135	System components need lower-level explanation with sample hazards; authentication module lacks detail	Added changes to HA system boundaries, splitting section to remove ambiguity	4bd898a
Peer Review (Group 14)	#142	FMEA rows should represent detailed failure modes rather than effects	Included further decomposition of table within FMEA	Different commit
Peer Review (Group 14)	#153	Technical hazards could also be safety hazards leading to unsafe exercise	Clarified hazard categorization	4381828
Peer Review (Group 14)	#152	Document lacks detailed mitigation strategies and implementation specifics	Mitigation included; will add section about effectiveness	None (different section)
TA	N/A	For Section 8 (Roadmap), including rationale would be helpful for which safety requirements to implement and why they are in that time frame	Updated the section with timelines and rationale for why each safety requirement was placed in each term	Similar to #144

Table 1: Feedback and changes for SRS and Hazard Analysis

1.2 Design and Design Documentation

1.3 VnV Plan and Report

2 Extras

[Summarize the extras that were tackled by this project. Extras can include usability testing, code walkthroughs, user documentation, formal proof, Gender-Mag personas, Design Thinking, etc. Extras should have already been approved by the course instructor as included in your problem statement. —TPLT]

3 Design Iteration (LO11 (PrototypeIterate))

[Explain how you arrived at your final design and implementation. How did the design evolve from the first version to the final version? —TPLT]

[Don't just say what you changed, say why you changed it. The needs of the client should be part of the explanation. For example, if you made changes in response to usability testing, explain what the testing found and what changes it led to. —TPLT]

4 Design Decisions (LO12)

[Reflect and justify your design decisions. How did limitations, assumptions, and constraints influence your decisions? Discuss each of these separately. —TPLT]

5 Economic Considerations (LO23)

[Is there a market for your product? What would be involved in marketing your product? What is your estimate of the cost to produce a version that you could sell? What would you charge for your product? How many units would you have to sell to make money? If your product isn't something that would be sold, like an open source project, how would you go about attracting users? How many potential users currently exist? —TPLT]

6 Reflection on Project Management (LO24)

[This question focuses on processes and tools used for project management. —TPLT]

6.1 How Does Your Project Management Compare to Your Development Plan

[Did you follow your Development plan, with respect to the team meeting plan, team communication plan, team member roles and workflow plan. Did you use

the technology you planned on using? —TPLT]

6.2 What Went Well?

[What went well for your project management in terms of processes and technology? —TPLT]

6.3 What Went Wrong?

[What went wrong in terms of processes and technology? —TPLT]

6.4 What Would you Do Differently Next Time?

[What will you do differently for your next project? —TPLT]

7 Ethics and Equity (LO18)

[Describe how your team applied the principle of equity and universal design to ensure equitable treatment of all stakeholders. —TPLT]

8 Reflection on Capstone

[This question focuses on what you learned during the course of the capstone project. —TPLT]

8.1 Which Courses Were Relevant

[Which of the courses you have taken were relevant for the capstone project? —TPLT]

8.2 Knowledge/Skills Outside of Courses

[What skills/knowledge did you need to acquire for your capstone project that was outside of the courses you took? —TPLT]